

Wireless Access Electromagnetics

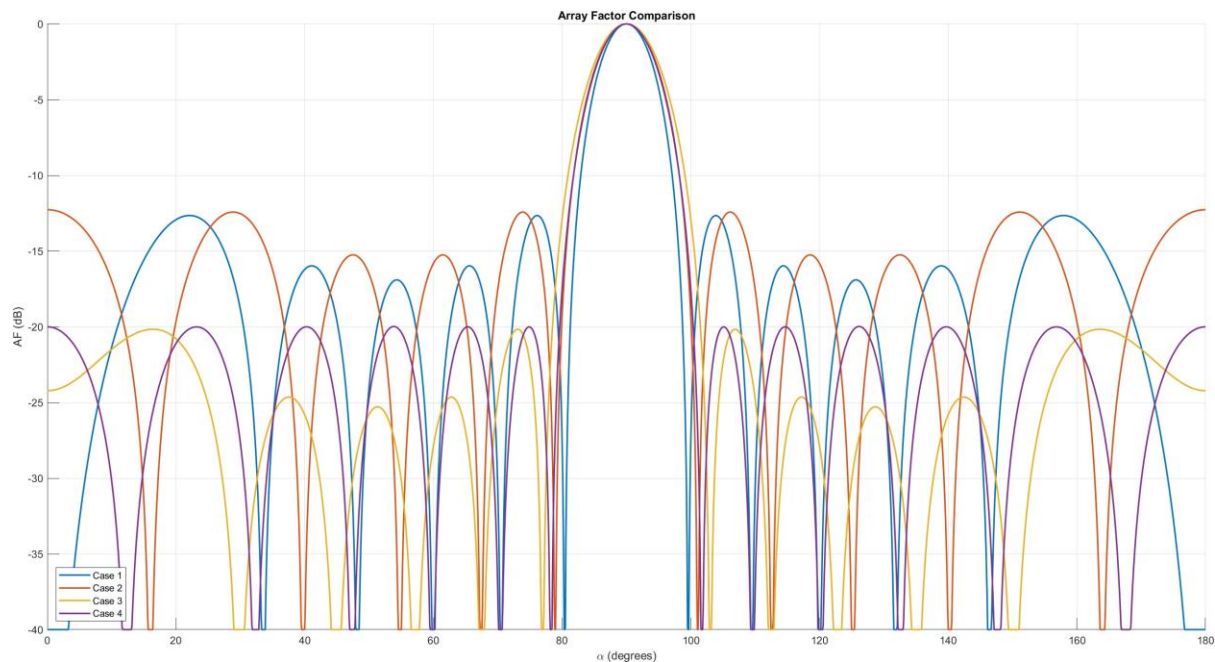
Homework 5

Sheharyar Ahmed Nizamani
S329580

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Problem 1:

Array factor comparison for 4 different cases:



Results & Discussion:

I plotted the array factors for the four different configurations, all showing a main lobe at 90° corresponding to broadside radiation. The two uniform arrays (Cases 1 and 2) exhibit sidelobe levels around -13 dB, as expected for uniform excitation. The non-uniform arrays (Cases 3 and 4) achieve sidelobe suppression below -20 dB through amplitude tapering, at the cost of the main beam being a bit wider. Case 4 provides the best suppression; it confirms that increasing the taper strength improves sidelobe.

Problem 2:

Broadside array design alongside x, and radiation patterns of the antenna in the 3 planes:

Nulls calculation: $\psi = \pm \frac{2m\pi}{N}$ $1 \leq m \leq N-1$

$$\cos(\alpha) = \pm \frac{2m\pi/N}{k \cdot d} \quad \because k \cdot d = 2\pi \frac{d}{\lambda} = 2\pi \Rightarrow 5 \times 0.8 = 4$$

$$\cos \alpha = \pm \frac{m}{4}$$

$$\text{For } m=1 \rightarrow \psi_1 = \frac{2\pi}{5} \Rightarrow \alpha_1 = 75.5^\circ$$

$$\rightarrow \psi_{-1} = \frac{2\pi}{5} \Rightarrow \frac{\psi_{-1}}{2\pi} = -0.2 \rightarrow \alpha_{-1} = 104.48^\circ$$

$$m=2 \rightarrow \psi_2 = \frac{4\pi}{5} \Rightarrow \alpha_2 = 60^\circ$$

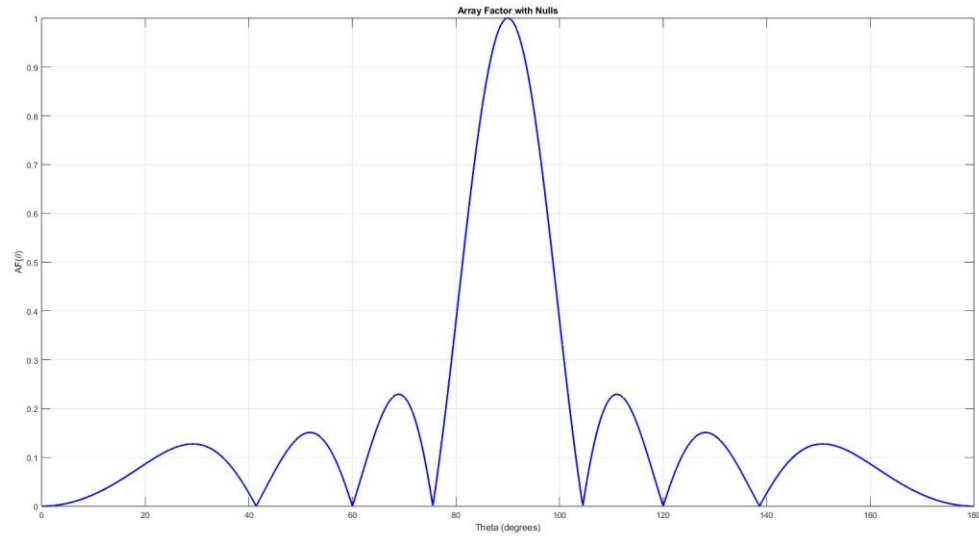
$$\rightarrow \psi_{-2} = \frac{4\pi}{5} \Rightarrow \frac{\psi_{-2}}{2\pi} = -0.4 = 120^\circ$$

$$m=3 \rightarrow \psi_3 = \frac{6\pi}{5} = \psi_3/2\pi = 0.6 \Rightarrow \alpha_3 = 41.4^\circ$$

$$\rightarrow \psi_{-3} = \frac{6\pi}{5} \Rightarrow \frac{\psi_{-3}}{2\pi} = -0.6 \Rightarrow \alpha_{-3} = 138.6^\circ$$

$$m=4 \rightarrow \psi_4 = \frac{8\pi}{5} \Rightarrow \frac{\psi_4}{2\pi} = 0.8 \Rightarrow \alpha_4 = 0^\circ$$

$$\rightarrow \psi_{-4} = \frac{8\pi}{5} \Rightarrow \frac{\psi_{-4}}{2\pi} = -0.8 \Rightarrow \alpha_{-4} = 180^\circ$$



I calculated the $N = 5$ and the distance between the elements was chosen to be equal to 0.8.

For a broadside array, the phase shift ϕ is zero to ensure that the radiation is directed perpendicularly to the array axis.

Radiation patterns:

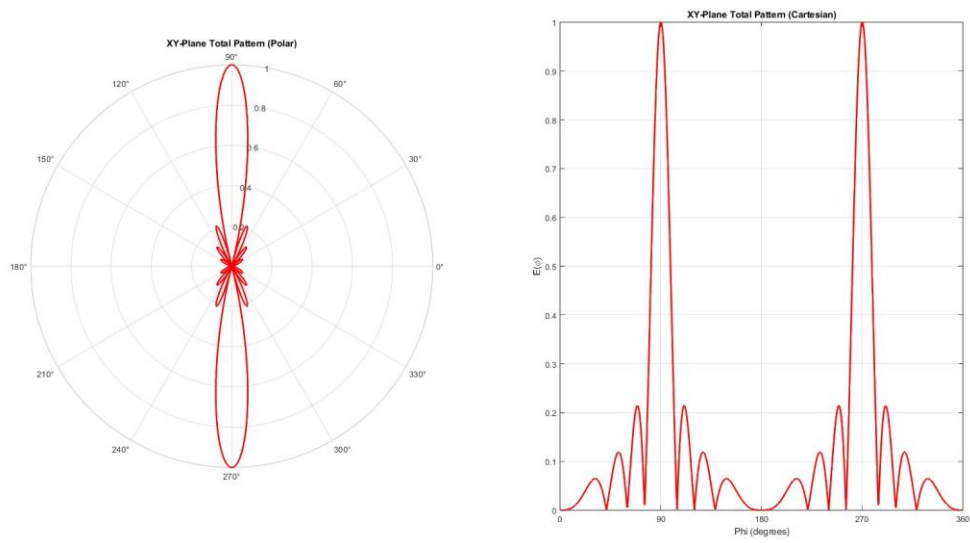


Fig 1: Radiation pattern in the xy plane

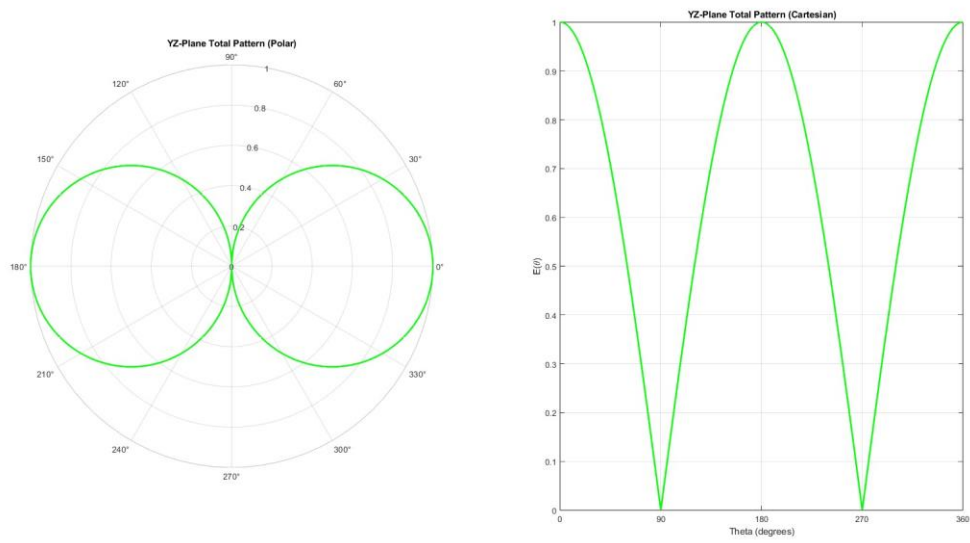


Fig 2: Radiation pattern in the yz plane

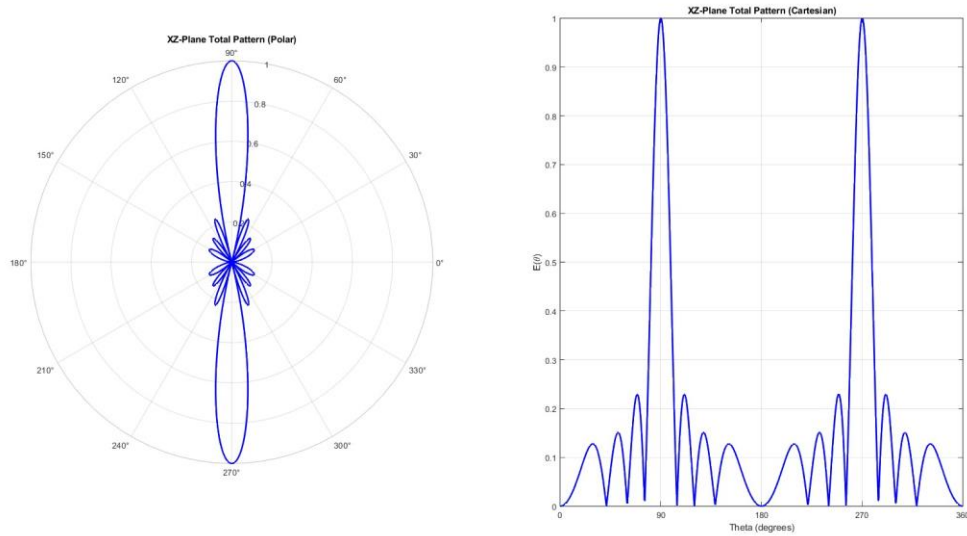


Fig 3: Radiation pattern in the xz plane

Results & Discussion:

The three plots show the total radiation pattern in each principal plane, combining the array factor with the dipole element pattern. In the XZ-plane, since the dipole along y is omnidirectional, the pattern is entirely shaped by the array factor; a narrow broadside beam at 90° with sidelobes on both sides. The XY-plane looks similar, with the dipole having minimal effect in this plane. The YZ-plane is different, here the array factor is constant, so the pattern is just the dipole figure-8 shape, with two lobes along y and nulls at 90° and 270° , exactly as expected for a y-directed dipole.

Problem 3:

Broadside array design alongside z, and radiation patterns of the antenna in the 3 planes:

Nulls: $\psi = -\frac{2m\pi}{N}$ $1 \leq m \leq N-1$

$k_d(\cos\alpha - 1) = -\frac{2m\pi}{7} \Rightarrow \cos\alpha - 1 = \frac{-2m\pi}{7k_d} = -\frac{m}{3}$

$\cos\alpha = -\frac{m}{3} + 1$

$m=1, \alpha_1 = 48.2^\circ$

$m=2, \alpha_2 = 70.5^\circ$

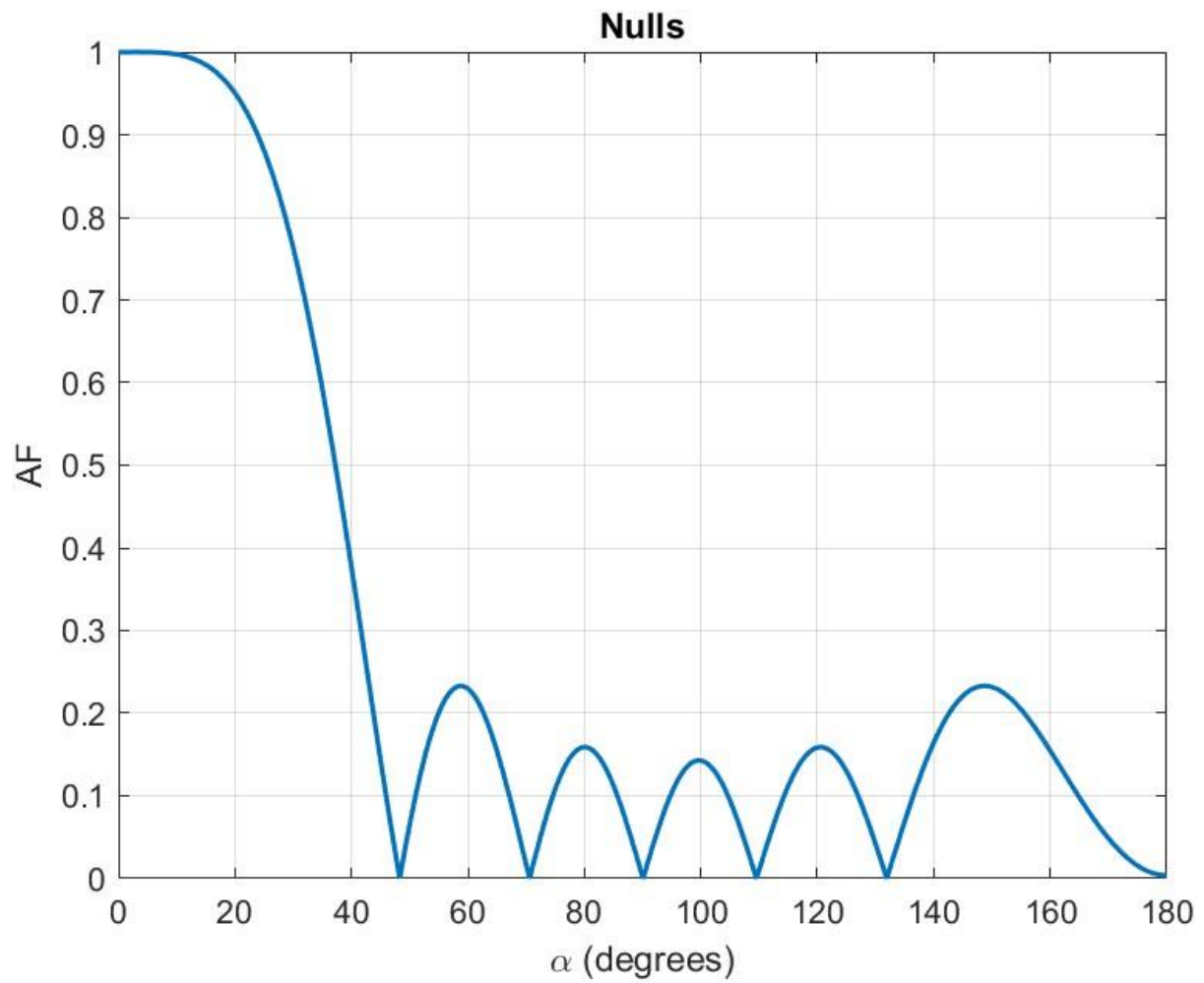
$m=3, \alpha_3 = 90^\circ$

$m=4, \alpha_4 = 109.5^\circ$

$m=5, \alpha_5 = 131.8^\circ$

$m=6, \alpha_6 = 180^\circ$

CS Scanned with CamScanner



I calculated the $N = 7$ and the distance between the elements was chosen to be equal to 0.4286. The phase shift was equal to -2.69 rads.

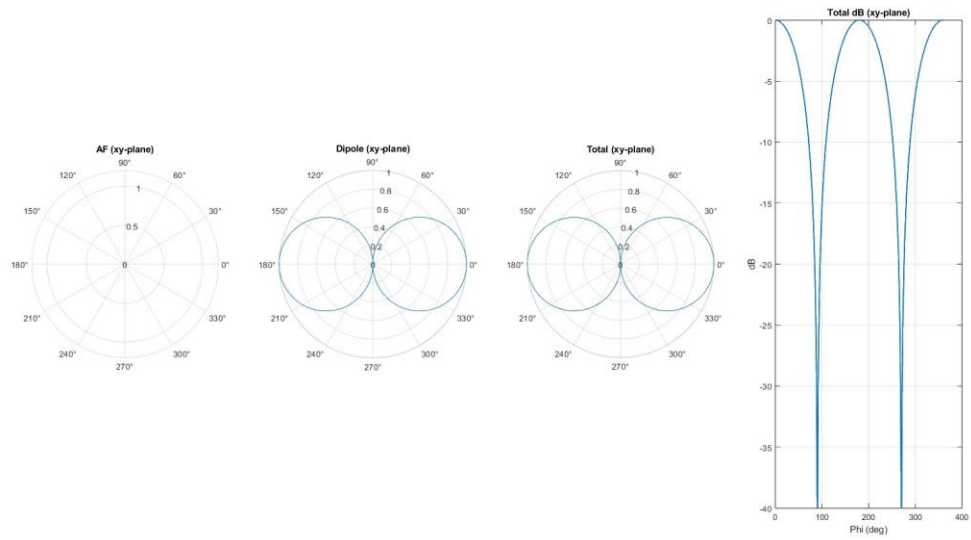


Fig 4: Radiation pattern in the xy plane

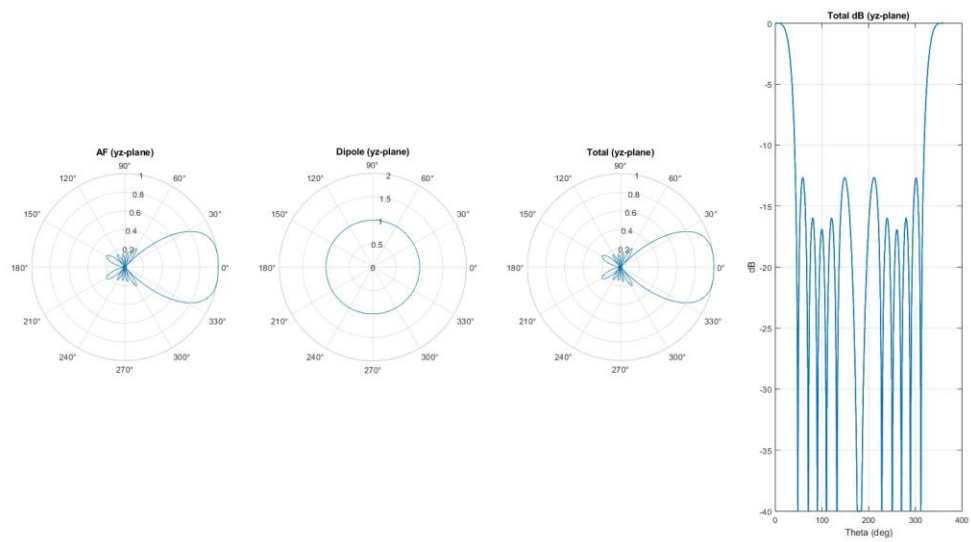


Fig 5: Radiation pattern in the yz plane

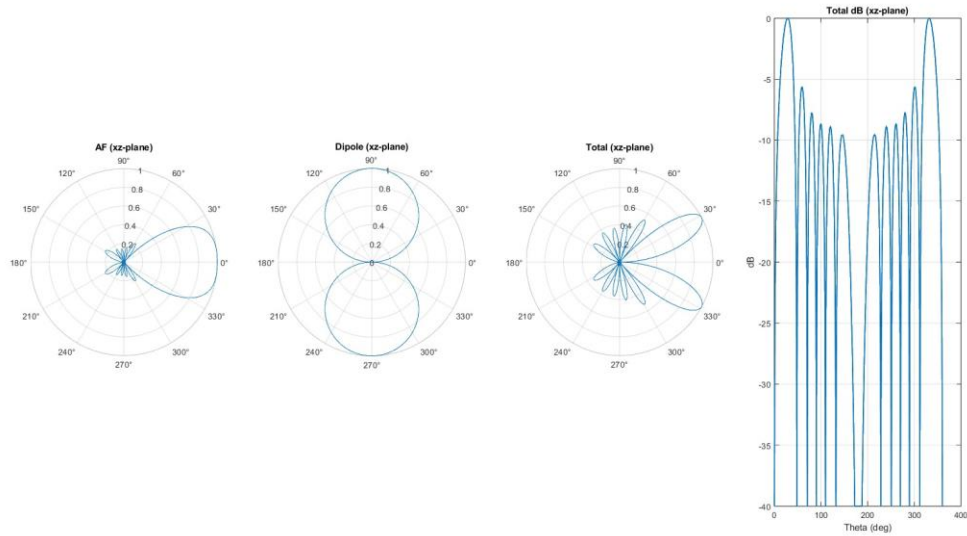


Fig 6: Radiation pattern in the xz plane

Results & Discussion:

The three principal plane patterns confirm the expected behaviour of the endfire array with x-directed dipole elements.

In the YZ-plane, the dipole is omnidirectional, so the total pattern matches the array factor directly, showing a clear endfire beam at 0° with visible sidelobes spreading across the visible region. In the XZ-plane, the dipole figure-8 null along x suppresses the endfire beam, redistributing energy toward broadside and producing a more complex multi-lobe total pattern. In the XY-plane, the array contributes no directivity since it points along z, so the total pattern is just the dipole figure-8 with lobes along y and deep nulls along x, confirming that in this plane the element pattern completely dominates.

Problem 4:

Ex 4: $f_0 = 5.8 \text{ GHz}$, $R_{in} = 119.6 \Omega$, $Z_0 = 50 \Omega$, $\epsilon_r = 4.45$, $h = 1.55$

Sol: $Z_B = Z_C = \frac{R_{in}}{2} = \frac{119.6}{2} = 59.8 \Omega$

$$Z_A = \frac{Z_{\infty_1}}{2} \Rightarrow Z_{\infty_1} = 2Z_A \quad Z_A = Z_0$$

At $R_{in} = 119.6 \Omega$, $W = 0.395 > 0.15$, $l = 7.5$
 $\downarrow Z_{\infty_2}$

$$Z_{QWT_1} = \sqrt{Z_0 \cdot Z_{\infty_2}} = \sqrt{50 \times 119.6} = 77.3 \Omega$$

For Z_0 , $W = 2.29$, $l = 7.12$

$$Z_{QWT_1} = W \approx 1.28, l = 7.27$$

